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Posted by u/papadiche
Catalina - 10.15
2 months ago

i9-9900K | 64GB 3200 | RX 5600 XT | Silent iMac Pro Killer

SUCCESS

Hardware

- Intel i9-9900K @ 5.0GHz
- ASRock Z390 Phantom Gaming ITX/ac
- OLOy 64GB (2 x 32GB) DDR4 3200MHz MD4U323216DJDA
- Samsung 960 Pro M.2 (2 TB)
- Samsung 860 Evo

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SATA (2 TB)

- Samsung 860 Evo SATA (4 TB)
- Samsung 860 Evo SATA (500 GB)
- BCM94360CD Wi-Fi / Bluetooth
- Hobbypower NGFF(M.2) Key A/E Adapter
- Silverstone 650 W 80+ Gold Certified Fully Modular SFX
- Sapphire Pulse RX 5600 XT
- NCase M1

Pics

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Overview

Considering the high cost and low upgradability of professional Mac computers, I built my own! Long ago I outgrew my Late 2013 MacBook Pro, and this led me to build my second hackintosh ever. I went the Clover bootloader route, since Opencore was in alpha, but Clover "locks you into" a specific version of macOS. If you ever upgrade macOS, you must update Clover first otherwise you will (temporarily) brick your computer!

For my third hackintosh (this one), I went with Opencore v0.5.8 and macOS Catalina 10.15.4. Once I got the config.plist working, it was smooth sailing! The best part is others have said they can update their computers through the App Store / Software Update, just like a real Mac! Fantastic news, and after confirming the NVRAM is natively supported on my motherboard, I have full confidence that is true.

Things that don't work 100%:

- Thunderbolt 3: Works on

cold boot for whatever device is plugged in at boot; that device can also be hot-plugged afterwards and works after Sleep. This means if you plug in a TB3 dock when you boot, you can hot plug devices into the dock, and daisy chain just like a real Mac. The ASRock Z390 Phantom ITX board has a JHL6240. If you're not familiar with TB3, the JHL6240 chip is a half-speed, half-power, 2-lane PCIe implementation of the older Alpine Ridge generation. The new ASRock Z490 Phantom ITX has a full-speed, full-power, 4-lane PCIe implementation of the new Titan Ridge chip, the JHL7340. Others have reported that the JHL7340 allow perfect hot plug in-and-out of Sleep, without the need for a TB3 plugged in at cold boot. Furthermore, you can enable Thunderbolt 3 Local Node and Target Display mode if you flash the firmware on the JHL7340 chip. CaseySJ has a great thread about this on the tonymacx86 website.

- TL;DR: If you need properly working TB3, go with the new ASRock Z490 Phantom ITX board and the new Intel i9-10xxx series processors.

- FAT32-formatted USB storage devices do not reconnect after Sleep, even with the Jettison app installed.
- All other USB

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storage devices require the Jettison app installed to eject correctly. While I have the XMP profile enabled in the BIOS (this automatically clocks the RAM at its rated 3200MHz instead of the Intel-standard 2133MHz), this had no effect on USB storage devices ejecting uncorrected upon Sleep. I was only able to get USB storage devices to eject properly when the RAM was set to the XMP profile, and clock-capped at 1600MHz with 1.4v of power. Using any of the standard, stock RAM profiles resulted in incorrect ejection, as did the unedited XMP profile.

- TL;DR: Install and use the Jettison app so you never have to worry about USB storage devices ejecting incorrectly upon Sleep.

Things that work 100%:

- Sleep (including Power Nap)
- USB ejection and remounting (with Jettison app installed)
- Thunderbolt 3 + TB3 Dock
- USB 3.0 Ports
- Wi-Fi

- Bluetooth
- iCloud
- iMessage
- AirDrop
- Continuity
- Handoff
- Dark Mode
- Find My Mac

Overclock

Considering the 9900k is notorious for being an overclock-able CPU, I tried every combination imaginable with Fixed Voltage and eventually wound up with the highest scores and best thermals with the following settings changed from stock:

```
> Core Ratio : 50
```

```
> Cache Rat
io : 46
> BCLK : 10
0
> FCLK : 1G
Hz
> CPU Tjunc
tion Max :
100
> Max Long
Power : 409
5
> Max Short
 Power : 409
> Max Amps:
 255
> XMP Profi
le 1
> 3200MHz D
RAM (should
 be set auto
> VCore : -
50 Offset
> LLC : Lev
el 3
> VCCIO : 1
.12
> VCCSA : 1
.12
```

These settings raised the CPU's Cinebench R20 scores by 16%, and the CPU's Geekbench 5 scores by 6%. Thermals were reduced by about 5C over stock in most working scenarios.

Temperatures

CPU temperatures are better than any other case I've used! If pushed for extended periods of time, the CPU can hit 85C with the fans running at around 80% speed. For shorter bursts of processing, such as rendering a song out of Logic Pro X, the CPU temperature will top out at ~70C. Over 15 minute hard processing, such as cryptomining or CPU-based video rendering, 75C is common. I have yet to see the CPU hit 90C even for short durations of time.

GPU temperatures are not reported on macOS, but I was able to check them with a Live CD. I can confirm it does not thermal throttle even on very difficult benchmarks nor playing video games at 4K. Gaming at 4K (performed on macOS) results in many frame drops, but again, no degradation in performance over extended periods of time, showing that it is not thermal throttling and instead just hitting its processing limit. Similar results are shown if rendering 4K video over and over again; the render times remain consistent. These tests were performed using the stock GPU fans, and after they were removed and replaced with two fans mounted to the bottom of the case. Identical performance. Though macOS has no way to currently check GPU temperatures, I'm confident the thermal performance is identical if not improved by using the larger, higher CFM airflow fans.

Noise

In the BIOS, I set up identical custom fan profiles for the

"CPU FAN," "CPU OPT," and "CHASSIS" outputs. In my case, I have the "CPU FAN" connected to the two Noctua NF-A15 fans on the Kraken X53, the "CPU OPT" connected to the Kraken X53 pump and the rear Noctua NF-A9 exhaust fan, and the "CHASSIS" connected to the Noctua NF-A15 and Kraken X53 120mm bottom exhaust fans. The custom fan profile is as follows:

```
> Temperature 1 : 20
> Fan 1 % : 40
> Temperature 2 : 50
> Fan 2 % : 50
> Temperature 3 : 60
> Fan 3 % : 60
> Temperatu
```

```
re 4 : 70
> Fan 4 % :
 70
> Critical
Temperature
 : 80
```

This fan profile provides the same thermals as the Performance settings but at lower noise levels than the Silent settings.

At idle, the fans are whisper quiet at around 33dB. For normal 50% CPU loads, the fans spin up to around 35dB. Under difficult process loads, the fans spin as loud as 41dB, and for absolutely all-out maximum 100% CPU loads, the fans get up to 45dB. Unless you are cryptomining or rendering long movies/videos through the CPU, your fans will stay in the "whisper quiet" to "reasonably quiet" range. Pushing the computer to be "loud" was something I did simply to ensure it would typically stay quiet, and to verify how loud it would get if somehow pushed to maximum loads.

Geekbench 5 Benchmarks

- Intel i9-9900k: 1370 Single Core | 9300 Multi-Core
- Sapphire Pulse RX 5600 XT: 62100 for Metal | 53000 for OpenCL

Cinebench R20 Benchmarks

- Intel i9-9900k: 5150

Build Notes

You can fit only 1x 2.5" SSD on the front panel using the standard NCase M1 screws and brackets. In my case, I fit 3x 2.5" SSDs by using double-sided tape (with small cardboard spacers to allow the SATA power connectors to fit correctly); they are mounted by tape to the "inside" of the PSU. This was the only way I could find to fit all three drives and the AIO in the case with reasonable cable management.

Wi-Fi antennas are simply on top of the cables at the ceiling of the case. The wires and connections are very stable if electrical tape is used on the BCM94360CD Wi-Fi chip to keep the MHF2 antenna connectors securely connected. You can also double-side tape the antennas to the side railings if desired, but I haven't found the need thus far; the antennas don't even move if the case is shaken. Do note that the BCM94360CD uses the larger MHF2 antenna connector, whereas the typical BCM94360CS2 Wi-Fi chip that most people purchase has the smaller MHF4 antenna connector. After trying both chips, the Wi-Fi performance seems nearly identical however the Bluetooth performance is much better with the BCM94360CD. The BCM94360CD has three Wi-Fi antennas and one Bluetooth antenna, whereas the BCM94360CS2 has two Wi-Fi antennas only. A lack of Bluetooth antenna on the BCM94360CS2 significantly hinders its Bluetooth performance.

The RAM clock primarily affects

the Geekbench 5 Multi-Core score. At 1600MHz and 1.4v, the Multi-Core score is 7300. At 2133MHz and 1.2v, the Multi-Core score is 8100. At 3200MHz and 1.35v (standard XMP Profile), the Multi-Core score is 8900.

The 240mm Kraken X53 AIO is a very, very tight fit in this case. There is at most a 2mm clearance length-wise with a Noctua NF-A9 PWM fan installed at the rear of the case. I significantly damaged the appearance of mine fitting it into the case. Many scratches were created!

Per OptimumTech's YouTube channel

suggestions, I have my fans oriented as follows: 2x Noctua NFA12x25 fans mounted on the AIO "pull" air through and exhaust onto the motherboard (this means the fan grills are pointed inwards with the radiator mounted closest to the outside). Stock GPU fans were removed, and instead a 140mm Noctua NF-A15 PWM and a Kraken X53 120mm fan was installed in a "pull" fashion, exhausting air down the bottom of the case (this means the fan grills are pointing down, against the bottom of the case). Both these fans are connected to the Chassis Fan pins on the motherboard, and set to monitor the CPU temperature. I wish I could set them to monitor the GPU temperature instead! Lastly, the PSU fan is facing the side of the case (instead of the inside), and the rear Noctua NF-A4 PWM fan is exhausting with its grill facing the back of the case. Overall, really solid thermals!

Logic Pro X Performance

350 Tracks with 1000+ Plugins and 500 Voices (from Virtual Instruments)

runs around 70% CPU load at 65C, and generates around 38dB of fan noise. Quiet enough to where the computer could be in the vocal booth, and minimal-to-zero noise would be heard through the microphone.

Install

Use the standard, up-to-date Opencore guide: https://dortania.github.io/OpenCore-Desktop-Guide/

My short-hand guide with settings specific to *this exact hardware configuration* can be viewed here: https://tinyurl.c om/y7cpz2ez

I installed the parts in this order:

1. Bottom exhaust fans
2. Power supply
3. Wi-Fi on Mobo (not in case)
4. M.2 drive on Mobo (not in case)
5. Motherboard (into case)
6. RAM
7. Front Panel wiring
8. Bottom Fan wiring
9. GPU
0. Rear fan
1. GPU wiring
2. SATA drives
3. Noctua fans on Kraken X53 (not in case)

4. Kraken X53 pump (into case)
5. Kraken X53 radiator
6. Wi-Fi Antennas

Improvements

- GPU Fan Control: If I had to do this build over again, I might power one or both of the bottom case fans from the GPU directly. This requires a "CRJ to 4-pin PWM adapter" as can be purchased here: https://www.amazon.

com/CRJ-4-Pin-Adapter-Sleeved-Graphics/dp/B07Q5BTTDX

- Wi-Fi + Bluetooth Antennas: After testing many varieties, I think a combination of 6dBi and 8 or 9dBi antennas is optimal for both near-range and long-range signals. My particular combination is 1x 6dBi (Wi-Fi) + 2x 8dBi (Wi-Fi) + 1x 8dBi (BT). When enclosed in the case, I can only get AirDrop to work from within 8 inches to the case, and Wi-Fi signals are only found up to ~80 linear feet away (including walls). If I had to do this build over again, I'd likely purchase 8dBi laptop antennas and mount the antenna outside the case, with wires going in through either the Wi-Fi holes or the water tube holes:

- BCM 9436 0CS2 (MHF

4 connector): https://www.amazon.com/Pair-53cm-20-86in-IPEX4-Antennas/dp/B07K68NHVH/ref=pd_sim_147_4/141-7940738-4854701?_encoding=UTF8&pd_rd_i=B07K68NHVH&p

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VBSJ
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- BCM94360CD (MHF2 connector): https://www.amazon.com/PORJET-Wireless-PCI-Bluetooth-Internal/dp/B01E211DLW/ref=pd_sim_147_7?

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- Spotlight: Typing in a math equation, even as simple as 100+200, results in a brief CPU temperature of 90C and brief CPU utilization of ~60%. I might try adjusting the fan curve so they don't spin up during these very brief periods, especially since the CPU always returns to ~35C within seconds of the crazy spike.

- Maybe more to come! :)

Also posted on : https://pcpartpicker.com/b/mGGcCJ

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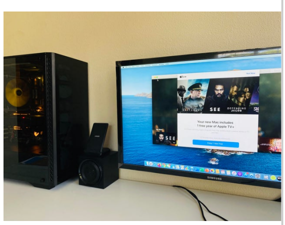

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